

**SIMATS ENGINEERING**  
**CO2\_ASSESSMENT TOOL3\_Application Based Questions**

**COURSE NAME:** Cloud Computing and Big Data Analytics      **COURSE CODE:** CSA1522

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|-----------------|------------------|
| Name            | <b>MAGESH S</b>  |
| Register Number | <b>192525002</b> |

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**COURSE OUTCOME COVERED**

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 25

Question Count: 5

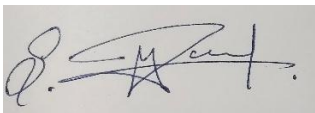
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**Code of Conduct**

I **MAGESH S (192525002)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



**Assessment Tasks**

**Application Based Questions**

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data?

Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

### Application Based Questions Rubric

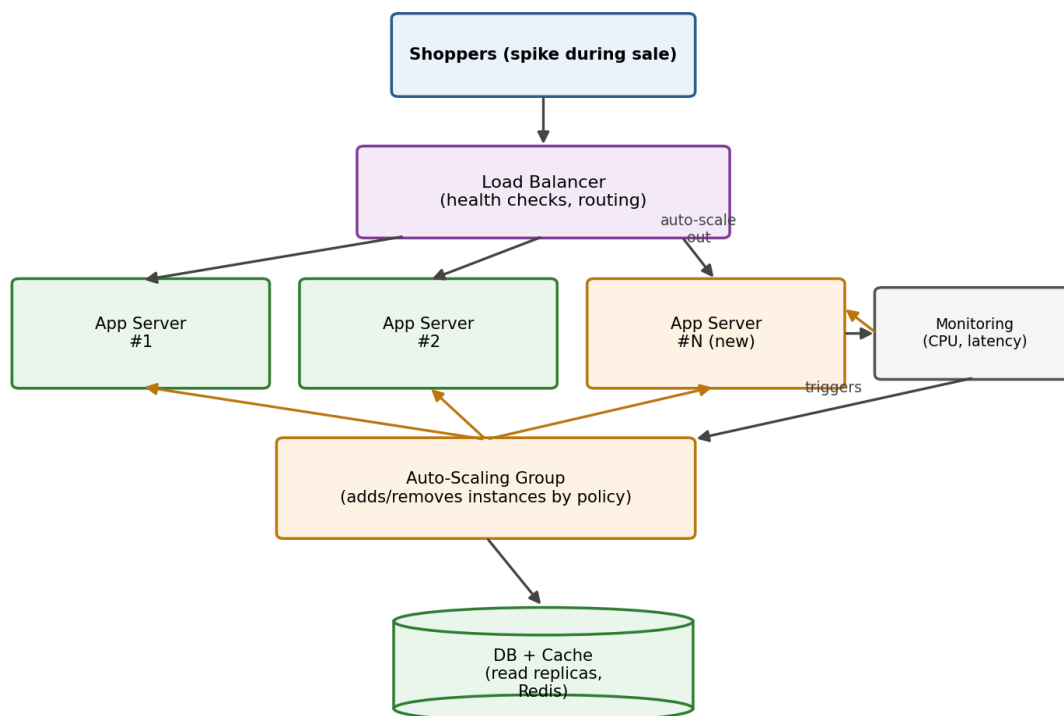
| Criteria                 | Weightage | Level 4 - Exemplary                                                  | Level 3 - Proficient                     | Level 2 - Developing         | Level 1 - Beginning     |
|--------------------------|-----------|----------------------------------------------------------------------|------------------------------------------|------------------------------|-------------------------|
| Relevance to Application | 25%       | Response directly addresses the given use case with strong relevance | Mostly relevant response                 | Partially relevant           | Weak relevance          |
| Use of Cloud Concepts    | 25%       | Applies appropriate concepts accurately                              | Mostly accurate application              | Basic application only       | Incorrect concept usage |
| Practical Insight        | 20%       | Shows strong real-world understanding                                | Shows reasonable practical understanding | Limited practical insight    | No practical insight    |
| Completeness             | 15%       | Response covers all required aspects                                 | Covers most aspects                      | Covers only basic aspects    | Incomplete response     |
| Clarity                  | 15%       | Clear and concise answer                                             | Generally clear                          | Some confusion in expression | Poor clarity            |

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

### How auto-scaling and load balancing handle unpredictable spikes

- Auto-scaling monitors real-time metrics (CPU utilization, request rate, queue length) and automatically adds compute instances/containers when demand rises, then removes them once traffic subsides — so capacity tracks actual demand instead of being fixed to a worst-case or average estimate.
- Load balancing distributes incoming requests evenly across all healthy instances, performing continuous health checks so traffic is never routed to an overloaded or failed server.
- Together, they let the platform absorb a sudden 10x traffic surge (e.g., a flash sale) by scaling out horizontally within minutes, and scale back in afterward to avoid paying for idle capacity.

#### Q1 — Auto-Scaling & Load Balancing for Traffic Spikes



### Reducing downtime and improving user experience

Because new instances spin up automatically before existing servers become saturated, response times stay consistent instead of degrading or timing out under load. Load balancer health checks route around failed instances instantly, so a single server crash doesn't take down the whole storefront — this directly reduces downtime and prevents cart-abandonment caused by slow or failed checkouts.

## **Role of monitoring tools**

Monitoring tools (CloudWatch, Prometheus/Grafana, Datadog) continuously track metrics like CPU, memory, latency, and error rate, feeding the data that triggers auto-scaling decisions. They also provide alerting so the operations team is notified of anomalies (e.g., a scaling limit being reached, a backend dependency slowing down) before customers are significantly affected — enabling proactive rather than reactive response.

## **Real-world e-commerce scenario**

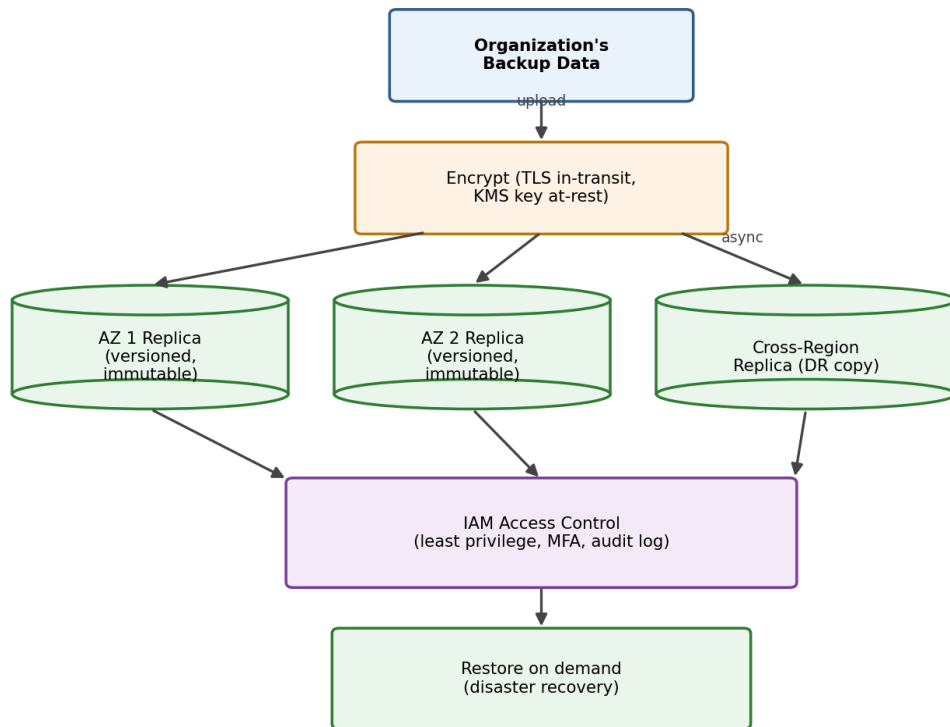
During a Black Friday / seasonal sale, an e-commerce platform like Amazon or Flipkart sees traffic jump many times over normal levels within minutes of a sale starting. Auto-scaling groups behind an Application Load Balancer add web/app server instances as request volume climbs, while the database tier uses read replicas and caching (e.g., Redis) to absorb read-heavy product-browsing traffic. Monitoring dashboards track checkout latency and error rates in real time, allowing engineers to catch and resolve bottlenecks (e.g., a payment gateway slowdown) within the sale window rather than after it has caused lost revenue.

- 2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.**

### **Durability and data redundancy**

- Cloud object storage (e.g., Amazon S3, Azure Blob Storage) automatically replicates each object across multiple physical devices and, often, multiple availability zones, achieving durability figures like "11 nines" (99.999999999%) — meaning the probability of losing a given object is vanishingly small.
- Cross-region replication further protects against a full regional disaster, keeping a synchronized copy of backup data in a geographically distant location.
- Erasure coding and checksums are used internally to detect and self-heal corrupted or lost data blocks without waiting for a full restore.

## Q2 — Secure Cloud Backup: Durability & Redundancy



### Encryption and access control for sensitive backup data

- Encryption at rest (using provider-managed or customer-managed keys via a KMS) ensures that even if underlying storage media were somehow accessed, the backup data remains unreadable without the key.
- Encryption in transit (TLS) protects backup data while it's being uploaded or restored over the network.
- Access control — IAM policies, bucket/container-level permissions, and the principle of least privilege — restricts who and what (users, applications, automation) can read, write, or delete backups.
- Immutable/WORM (write-once-read-many) storage and versioning protect backups from accidental or malicious deletion/modification, which is especially important as a defense against ransomware.

### Cost advantages compared to traditional storage

- No upfront hardware capital expenditure — no need to purchase and maintain tape libraries, disk arrays, or off-site storage facilities.
- Pay-as-you-go pricing means the organization only pays for the storage actually consumed, scaling costs directly with data growth.
- Storage tiering (hot, cool/infrequent-access, archive/glacier tiers) lets older backups automatically move to far cheaper storage classes, something traditional on-prem storage cannot do dynamically.
- Eliminates the operational cost of managing, cooling, and physically securing on-premises backup infrastructure.

## Disaster recovery use case example

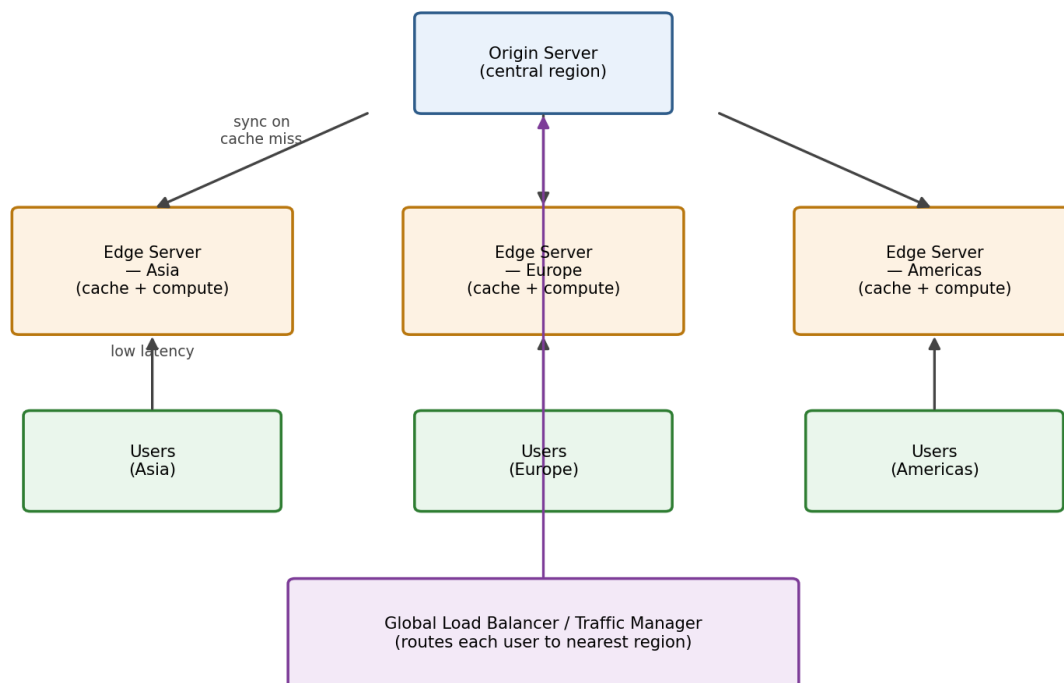
A hospital's patient records system backs up its database nightly to cloud object storage with cross-region replication. When a ransomware attack encrypts the primary on-premises database, the organization restores from the most recent immutable, versioned cloud backup — recovering clean data within hours instead of paying a ransom or permanently losing patient records, because the backup could not be altered or deleted by the attacker.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

## How CDNs improve performance and reduce latency

A CDN caches static and semi-static content (images, videos, app assets, API responses) on a globally distributed network of edge servers (Points of Presence), so a user's request is served from the nearest location instead of traveling back to a single origin server, which may be thousands of kilometers away. This reduces round-trip time (RTT), offloads traffic from the origin server (reducing origin load and cost), and improves resilience since traffic can be rerouted around a failed edge node.

### Q3 — Global CDN & Edge Server Architecture



## **How edge servers enhance user experience across regions**

Edge servers bring computation and data physically closer to end users, cutting the network latency that dominates response time for geographically distant users. Many CDNs now support edge compute allowing logic like authentication, A/B testing, or personalization to run at the edge instead of round-tripping to a central region — further reducing perceived latency. Regional edge caching also helps smooth out spikes in specific without impacting users elsewhere.

## **How cloud providers support global availability**

- Cloud providers offer a global network of regions and availability zones, letting companies deploy application instances close to major user populations.
- Global load balancing / traffic management (e.g., AWS Global Accelerator, Azure Front Door) routes each user to the nearest healthy regional deployment automatically.
- Multi-region data replication keeps user data consistent and available even if one region experiences an outage.

## **Real-time streaming/gaming example**

In a multiplayer mobile game, low latency is critical for a responsive experience — even 100ms of extra lag can visibly affect gameplay. The game company deploys game servers across multiple cloud regions and uses a CDN plus edge compute to handle matchmaking requests and static asset delivery (textures, updates) from the nearest edge location. Players are automatically routed to the nearest regional game server cluster via global load balancing, keeping in-game latency low regardless of whether the player is in India, Europe, or the US — directly improving retention and user experience.

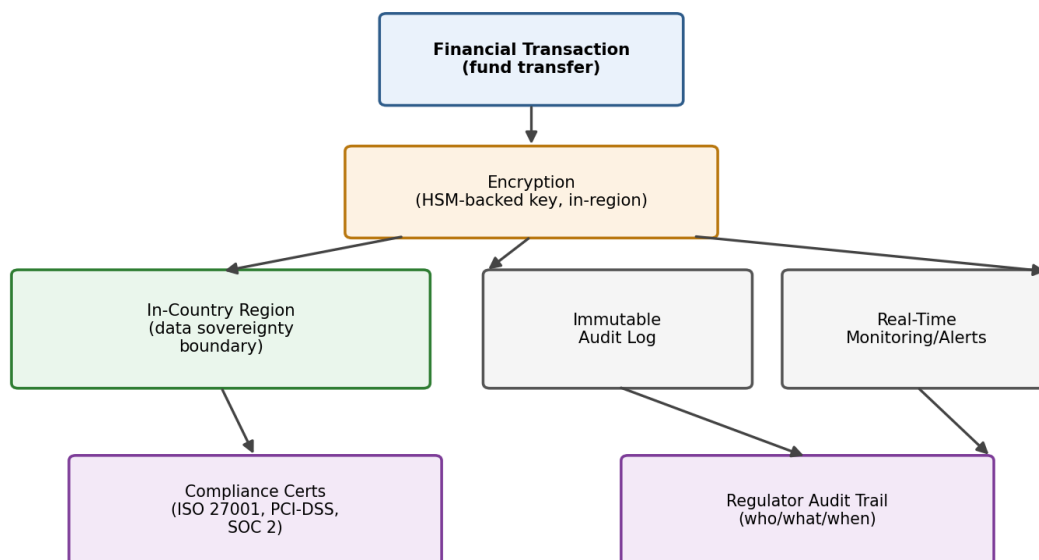
4. **A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.**

## **How cloud providers support regulatory compliance and auditing**

- Major cloud providers maintain compliance certifications (ISO 27001, SOC 2, PCI-DSS, GDPR-readiness, RBI/regional financial regulations) and publish audit reports, giving customers a documented baseline they can build on rather than proving everything themselves.
- Providers offer compliance-focused services — audit trail tools (AWS CloudTrail, Azure Monitor), configuration compliance scanners (AWS Config, Azure Policy), and dedicated compliance/financial-services regions with additional contractual and technical controls.
- Shared responsibility model: the provider secures the underlying infrastructure, while the institution remains responsible for correctly configuring identity, encryption, and access control on top of it — compliance requires both sides to do their part.



## Q4 — Financial Compliance & Security Architecture



### Role of logging, monitoring, and encryption in security

- Logging (API calls, access logs, transaction logs) creates an immutable, timestamped record of who did what, which is essential for forensic investigation and regulatory audit requirements.
- Monitoring continuously analyzes logs and metrics for suspicious activity (e.g., unusual access patterns, privilege escalation attempts) and triggers real-time alerts, enabling rapid response to potential breaches.
- Encryption (at rest and in transit) ensures that even if data is intercepted or storage is improperly accessed, transaction data and customer financial information remain unreadable without the proper keys — a baseline requirement for financial data under most regulations.

### Ensuring data sovereignty requirements

- Select a cloud region located within the required jurisdiction (e.g., a region physically inside the country whose laws mandate data residency) and explicitly restrict data storage and processing to that region.
- Use customer-managed encryption keys (CMEK) kept under the institution's control, rather than fully provider-managed keys, so data cannot be decrypted outside the sanctioned jurisdiction even by the provider.
- Contractually confirm with the provider where backups, replicas, and support access originate from, since sovereignty can be violated by cross-border replication or support staff accessing data from another country.

### Example involving sensitive financial transactions

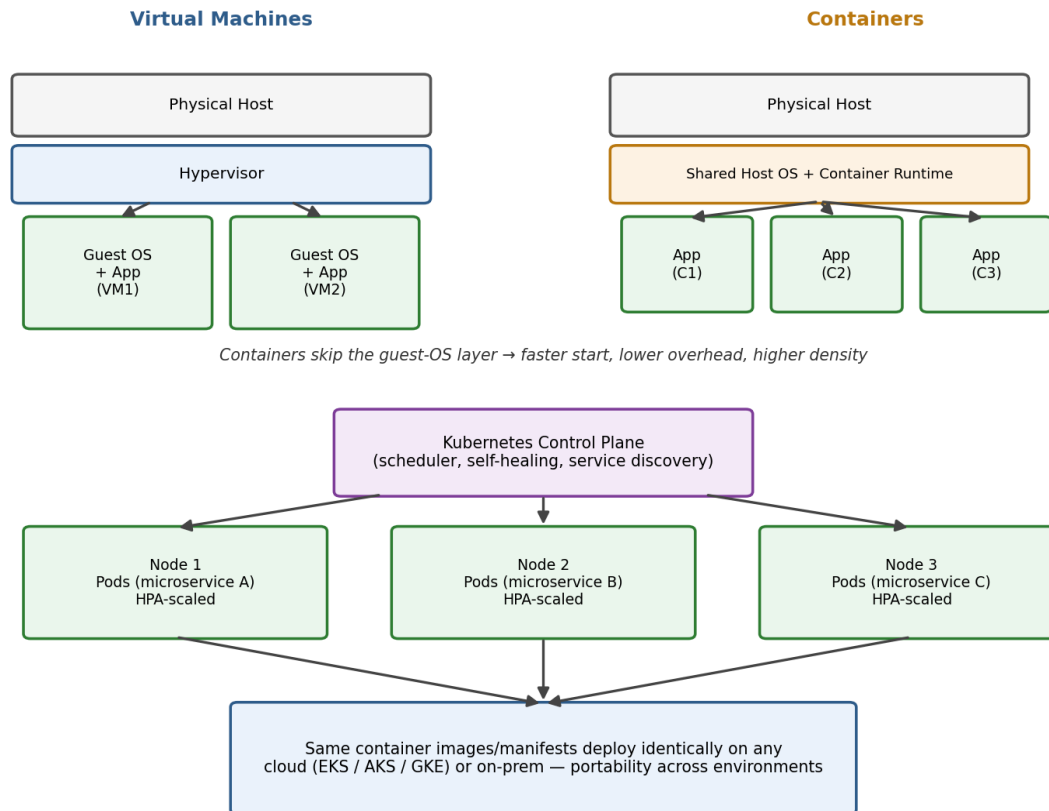
A bank processing online fund transfers encrypts each transaction record at rest using a customer-managed key stored in a regional Hardware Security Module (HSM), logs every transaction and administrative action to an immutable audit trail, and restricts all transaction data to cloud infrastructure located within the country per central bank data-localization rules. If a regulator later needs to audit a disputed transaction, the bank can produce a complete, tamper-evident log trail showing exactly when and by whom the transaction was processed and accessed.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

### Containers vs. virtual machines — performance and resource usage

| Aspect             | Virtual Machines                                         | Containers                                                                            |
|--------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| What's virtualized | Full hardware, including a complete guest OS per VM      | The OS kernel is shared; only the application and its dependencies are packaged       |
| Startup time       | Minutes (booting a full OS)                              | Seconds (just starting a process)                                                     |
| Resource overhead  | Higher — each VM carries its own OS memory/CPU footprint | Lower — no duplicate OS per instance, so far more containers fit on the same hardware |
| Isolation strength | Strong (hardware-level isolation via hypervisor)         | Lighter (process/namespace-level isolation via the OS kernel)                         |
| Density            | Fewer instances per physical host                        | Many more instances per physical host                                                 |

## Q5 — Containers vs VMs & Kubernetes Orchestration



### How orchestration (Kubernetes) simplifies deployment and scaling

- Kubernetes automates scheduling (placing containers on available nodes based on resource needs), self-healing (automatically restarting or rescheduling failed containers), and service discovery (containers find each other via stable internal DNS names regardless of which node they're running on).
- Horizontal Pod Autoscaling automatically increases or decreases the number of running container replicas based on load, without manual intervention.
- Rolling updates and rollbacks let the team deploy new versions gradually and safely revert if something goes wrong, with zero downtime.
- Declarative configuration (YAML manifests) means the desired state of the whole application is version-controlled and reproducible, rather than manually configured server by server.

### Portability benefits across cloud environments

A container image bundles the application and all its dependencies, so it runs identically on a developer's laptop, on-premises servers, or any major cloud provider — eliminating "it works on my machine" inconsistencies. Kubernetes itself is supported as a managed service across all major clouds (EKS, AKS, GKE) and on-prem, so the same deployment manifests can be applied in different environments with minimal changes, avoiding vendor lock-in and enabling multi-cloud or hybrid strategies.

### **Microservices-based application example**

An online food delivery app (e.g., a Swiggy/Zomato/DoorDash-style platform) is built as independent microservices — order management, restaurant catalog, payment processing, delivery-partner tracking, and notifications — each packaged as its own container. Kubernetes orchestrates all of these services, independently scaling the order-management and tracking services during peak lunch/dinner hours while keeping less-used services like account settings at baseline capacity, and allows the team to update the payment service without redeploying or risking downtime for the rest of the app.